Operating Costs	Probability
$\hat{Y}$ = 12.5 + 0.65(125) = USD93.75 million	0.80(0.50) = 0.40
$\hat{Y}$ = 12.5 + 0.65(100) = USD77.50 million	0.80(0.50) = 0.40
$\hat{Y}$ = 12.5 + 0.65(80) = USD64.50 million	0.20(0.85) = 0.17
$\hat{Y}$ = 12.5 + 0.65(70) = USD58.00 million	0.20(0.15) = 0.03
	Sum = 1.00

2. Compute the expected value of operating costs under the high growth scenario. Also calculate the expected value of operating costs under the low growth scenario.

Solution:

US dollar amounts are in millions.

$$\text{E(operating costs}|\textit{high growth}) = 0.50(\text{USD}93.75) + 0.50(\text{USD}77.50)$$

$$= \text{USD}85.625$$

$$\text{E(operating costs}|\textit{low growth}) = 0.85(\text{USD}64.50) + 0.15(\text{USD}58.00)$$

$$= \text{USD}63.525$$

3. Refer to the question in the initial box of the tree: What are BankCorp's expected operating costs?

Solution:

US dollar amounts are in millions.

$$E(\text{operating costs}) = E(\text{operating costs}|\textit{high growth})P(\textit{high growth})$$

$$+ E(\text{operatingcosts}|\textit{low growth})P(\textit{low growth})$$

$$= 85.625(0.80) + 63.525(0.20) = 81.205$$

BankCorp's expected operating costs are USD81.205 million.

In this section, we have treated random variables, such as EPS, as standalone quantities. We have not explored how descriptors, such as expected value and variance of EPS, may be functions of other random variables. Portfolio return is one random variable that is clearly a function of other random variables, the random returns on the individual securities in the portfolio. To analyze a portfolio's expected return and variance of return, we must understand that these quantities are a function of characteristics of the individual securities' returns. Looking at the variance of portfolio return, we see that the way individual security returns move together or covary is key. We cover portfolio expected return, variance of return, and importantly, covariance and correlation in a separate learning module.

QUESTION SET

1. Suppose the prospects for recovering principal for a defaulted bond issue depend on which of two economic scenarios prevails. Scenario 1 has probability 0.75 and will result in recovery of USD0.90 per USD1 principal value with probability 0.45, or in recovery of USD0.80 per USD1 principal